Definition 3: A function f is continuous from the right at a number a if

$$\lim_{x \to a^+} f(x) = f(a)$$

and f is continuous from the left at a if

$$\lim_{x \to a^-} f(x) = f(a)$$

Example 2: Determine the continuity of f at $x = 3$ and $x = -2$.

$$f(x) = \begin{cases} \frac{1}{x-2}, & x \leq -2 \\ e^x, & -2 < x < 3 \\ \ln(x+1), & x \geq 3 \end{cases}$$

Definition 4: A function f is continuous on an interval if it is continuous at every number in the interval. (If f is defined only on one side of an endpoint of the interval, we understand continuous at the endpoint to mean continuous from the right or from the left

Example 2 :

- $x=-2$

$$f(-2) = \frac{1}{-2-2} = -\frac{1}{4}$$

$$\lim_{x \to -2^-} f(x) = \lim_{x \to -2^-} \frac{1}{x-2} = -\frac{1}{4} = f(-2)$$

$$\lim_{x \to -2^+} f(x) = \lim_{x \to -2^+} e^x = e^{-2} \neq \frac{1}{4}$$

f is discontinuous at $x=-2$ but is left continuous at $x=-2$

- $x=3$

$$\lim_{x \to 3^-} f(x) = \lim_{x \to 3^-} e^x = e^3 \neq f(3) = \ln 4$$

$$\lim_{x \to 3^+} f(x) = \lim_{x \to 3^+} \ln(1+x) = \ln(1+3) = \ln 4$$

Hence f is right continuous at $x=3$ but discontinuous at $x=3$.

Theorem 1 : If f and g are continuous at a and c is a constant, then the following functions are also continuous at a:

1. $f+g$ 2. $f-g$ 3. cf

4. $f \cdot g$ 5. $\frac{f}{g}$ if $g(a) \neq 0$

Theorem 2 : the following types of functions are continuous at every number in their domains.

- polynomials
- rational functions
- root functions
- trigonometric functions
- exponential functions
- logarithmic functions
- inverse trigonometric functions

Theorem 3 : If f is continuous at b and $\lim_{x \to a} g(x) = b$, then $\lim_{x \to a} f(g(x)) = f(b)$.

In other words,

$$\lim_{x \to a} f(g(x)) = f\left(\lim_{x \to a} g(x)\right)$$

Theorem 4: If g is continuous at a and f is continuous at $g(a)$, then the composite function $f \circ g$ given by $(f \circ g)(x) = f(g(x))$ is continuous at a.

Example 3: Determine the intervals on which the functions are continuous

(a) $f(x) = \dfrac{x-1}{x+2}$ (b) $g(x) = \dfrac{\ln(3-x)}{\sqrt{e^x+1}}$

(c) $h(x) = \sin^{-1}\left(\dfrac{1}{x}\right)$

Example 4: Determine

(a) $\lim\limits_{x \to 1} \cos^{-1}(\ln x)$

(b) $\lim\limits_{x \to \frac{2}{3}^-} e^{\left(\sqrt{\frac{2}{3}-x}\right)}$

(c) $\lim\limits_{x \to 0^+} \sqrt{1-e^x}$

(d) $\lim\limits_{x \to 0^-} \sqrt{1-e^x}$

Example 3: Determine the intervals on which the functions are continuous.

(a) $f(x) = \frac{x-1}{x+2}$; (b) $g(x) = \frac{\ln(3-x)}{\sqrt{e^x+1}}$

(c) $h(x) = \sin^{-1}\left(\frac{1}{x}\right)$

Solution: (a) $D_f = (-\infty,-2)\cup(-2,+\infty)$

$\Rightarrow$ f is continuous on D_f since f is a rational function

(b) $D_g = \{x \in \mathbb{R} / 3-x > 0\}$ (since $e^x+1 > 0$)

$= (-\infty, 3)$

$\ln(3-x)$ is continuous on D_g

and $\frac{1}{\sqrt{e^x+1}}$ is continuous on $\mathbb{R}$ (in particular on D_g)

$\Rightarrow$ g is continuous on D_g

(c) $D_h = \{x \in \mathbb{R} / \left|\frac{1}{x}\right| \leq 1 \text{ and } x \neq 0\}$

$\left|\frac{1}{x}\right| \leq 1 \Rightarrow |x| \geq 1 \Rightarrow D_h = (-\infty, -1] \cup [1, +\infty)$

h is continuous on D_h

Example 4:

(a) $\lim_{x \to 1} \ln x = 0$ and $\cos^{-1}(x)$ is continuous for all $x \in [-1, 1]$. Therefore

$$\lim_{x \to 1} \cos^{-1}(\ln x) = \cos^{-1}(0) = \frac{\pi}{2}$$

(b) $\lim_{x \to \frac{2}{3}^-} \sqrt{\frac{2}{3} - x} = 0$ and e^x is continuous on $\mathbb{R}$ $\Rightarrow \lim_{x \to \frac{2}{3}^-} e^{\sqrt{\frac{2}{3} - x}} = e^0 = 1$

(c) $\lim_{x \to 0^+} e^x = 1 \ (1^+)$ ($\sqrt{1-x}$ is left continuous at $x = 1$)

but $\sqrt{1-x}$ is not right continuos at $x = 1$

$$\Rightarrow \lim_{x \to 0^+} \sqrt{1 - e^x} \neq \sqrt{1 - \lim_{x \to 0^+} e^x}$$

(Note that $\lim_{x \to 0} (1 - e^x) = 0^-$)

$\lim_{x \to 0^+} \sqrt{1-e^x}$ does not exist

(d) $\lim_{x \to 0^-} e^x = 1 \quad (1^-)$

and $\sqrt{1-x}$ is left continuous at $x = 1$

$$\Rightarrow \lim_{x \to 0^-} \sqrt{1-e^x} = \sqrt{1 - \lim_{x \to 0^-} e^x} = 0 = \left(1 - e^0\right)$$